Laboratory 11

Stopwatch Datapath and Control

11.1 Outcomes and Objectives

The outcome of this lab is complete the implementation of the stopwatch on the FPGA development board. Through this process you will achieve the following learning objectives.

- Datapath and Control Architecture
- Using a timing for a datapath and control circuit to specify or verify proper operation.
- Synthesizing a module on the FPGA development board

11.2 Stopwatch

From the previous lab, you should be familiar with the operation of our stopwatch. Briefly, our stopwatch allows a user to measure elapsed time and lap times of competitive events. Our stopwatch measures time in increments of a tenth of a second, unit second and tens of seconds. Control input comes from two buttons, called `S1` and `S2`, according to the finite state machine shown in Figure 11.1.

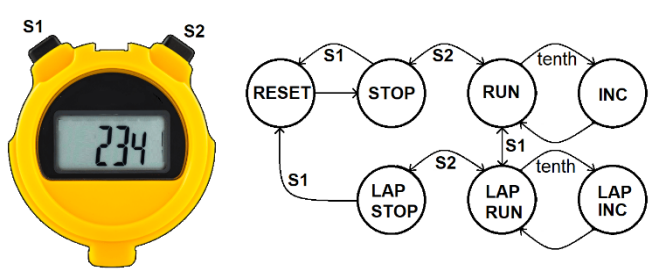

Figure 11.1: A digital stopwatch gets its input from two buttons and displays its output on a 7-segment display. The behavior of the stopwatch can be described by this finite state machine (FSM).

Figure 11.2 shows how you will implement the idea presented in Figure 11.1 using the FGPA development board.

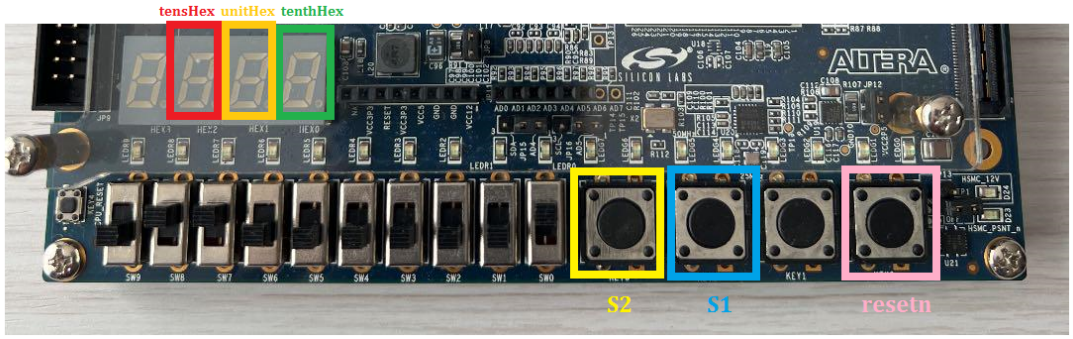

Figure 11.2: The stopwatch user interface as implemented on the FPGA development board.

Two of the buttons will act as the stopwatch buttons and three of the 7-segment displays will show the time. The `reset` signal is connected to a button and the 50MHz clock is on the FPGA development board but does not require any user interaction.

11.3 System Architecture

In the previous two labs you have created the datapath and control unit for the stopwatch. Using these two components as building blocks, the architecture for the stopwatch, shown in Figure 11.3, is almost trivial; the Verilog file for the stopwatch contains two component instantiations of the datapath and controlUnit.

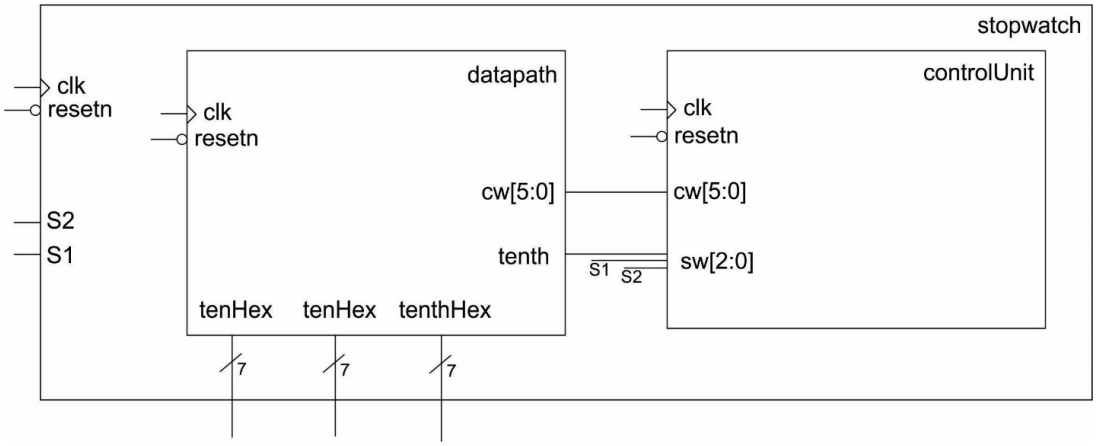

Figure 11.3: The architecture for the stopwatch consists of a datapath and controlUnit.

The only minor complication is combining the tenth output from the datapath with the `S2` and `S1` signals coming in from the buttons into a 3-bit signal sent to the status word input of the control unit.

Use the starter code provided on Canvas to complete the stopwatch module. While you are at it, download the stopwatch testbench provided on Canvas, and make the testbench the top-level entity.

11.4 Testbench

Before you download your completed control unit to the development boards, you are going to perform extensive simulations to uncover as many bugs as possible. Errors are much, much easier to find in a simulation. First you will need to understand what the testbench simulation is supposed to do.

The timing diagrams in Figure 11.4 show the `S1` and `S2` signals as they are manipulated by the testbench. The tenth signals will be asserted by your datapath but are included to help you. Your task is to fill in the symbolic name of the state that the FSM is in as well as the `clkCount` value, the `mod10Counter` outputs and the values displayed on the 7-segment displays.

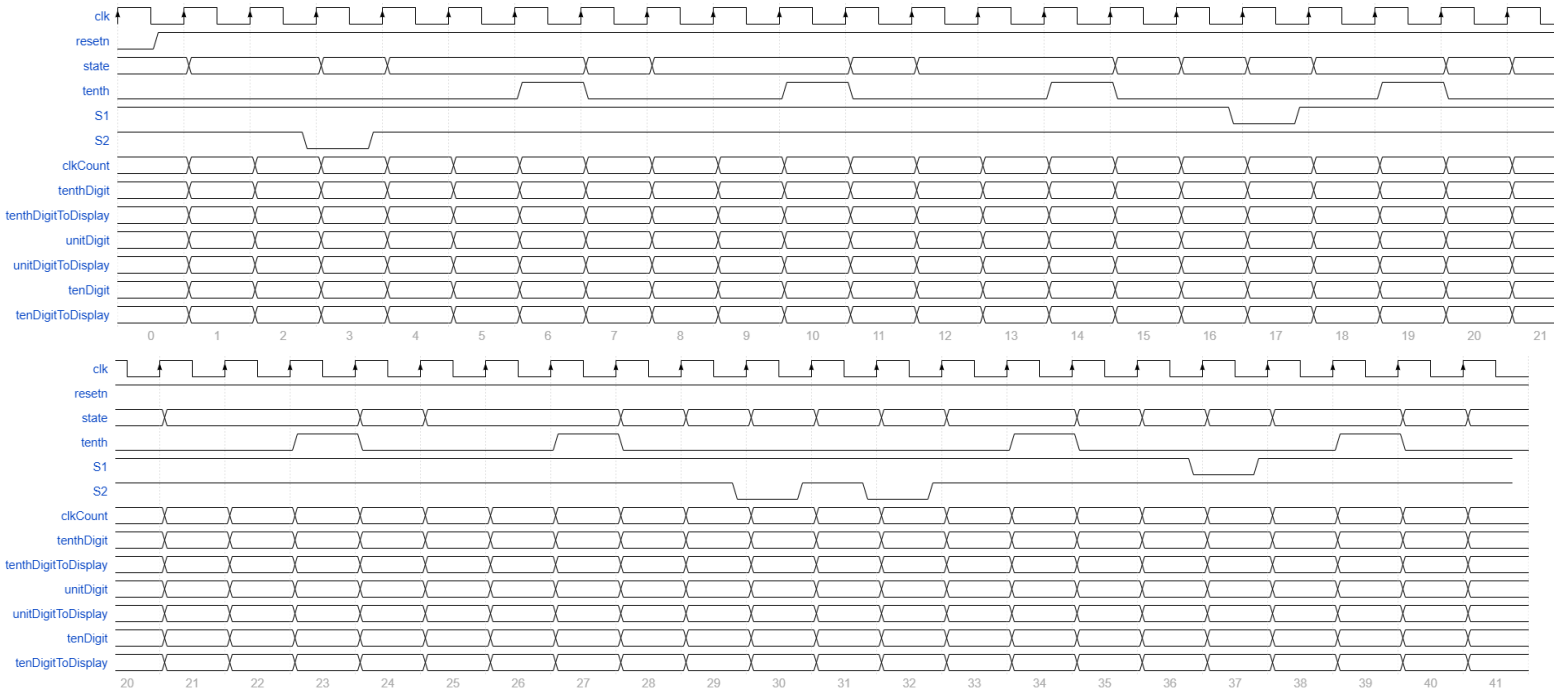

Figure 11.4: Timing diagram that you will run on the testbench.

Listing 11.1: Changes to the datapath that will allow it to run properly on the FPGA development board.

```
// parameter N = 4;
parameter N = 24;

// localparam    tenthSecondConstant    = 4'h000002;
localparam    tenthSecondConstant    = 24'h4c4b40;

//localparam    zero24 = 4'h000000;
localparam    zero24 = 24'h000000;
```

The goal of the testbench was to cover every transition arc in the state diagram. This goal was not achieved; several transition arc were not taken in the testbench, which one was it? I'd suggest double checking this transition during the testing of the stopwatch when it is downloaded onto the FPGA development board.

Run the testbench using the provided do file and compare the output against Figure 11.4 You will probably need to modify the do file to make it work with your design. Address any problems in the stopwatch before proceeding.

11.5 Pin-Assignment and Synthesis

When you created the datapath, you intentionally designed the timer counter to count up from 0 to 2 in order to expedite execution of the simulations. Before you synthesize the datapath, you need to undo this. I would suggest leaving the relevant constants in your code and just comment them out. Immediately following each commented constant, put the constant that you need for the datapath to operate correctly on the FPGA development board. I've summarized these changes in Listing 11.1.

While you are at it, make sure that you remove the testbench as the top-level module and make the stopwatch the top-level module. Then run the analysis and elaboration tool to make sure that the changes in Listing 11.1 did not create any warnings or errors

The next step is to create the mapping of stopwatch module inputs and outputs to the pins of the FPGA and by extension the input and output devices on the FPGA development board. Use the inputs and outputs shown in Figure 11.2 and the information in the FPGA development board User Guide to complete the following pin assignment tables.

Table 11.1: Pin assignment for the stopwatch.

S2	Key[3]	Y16
S1	Key[2]	
resetn	Key[0]	
clk	CLOCK_50	R20

Table 11.2: Seven-segment display pin assignments.

Segment	**tenHex (Hex2)**	**unitHex (Hex1)**	**tenthHex (Hex0)**
seg[6]			
seg[5]			
seg[4]	V20		
seg[3]			
seg[2]			V17
seg[1]			
seg[0]		AA18	

After making the pin assignment, download and test your design.

11.6 Turn in

You may work in teams of at most two. Make a record of your response to the items below and turn them in a single copy as your team's solution on Canvas using the instructions posted there. Include the names of both team members at the top of your solutions. Use complete English sentences to introduce what each of the following listed items (below) is and how it was derived. In addition to this submission, you will be expected to demonstrate your circuit at the beginning of your lab section next week.

Testbench

- Completed Figure 11.4. Please paste the images in landscape format into your solutions
- Screen shot of simulation timing diagram. Please compose the simulation in landscape format into your solutions and break the screen shot of the simulation into three parts,
  - from 0 to 400ps
  - from 400ps to 800ps
  - from 800ps to 1200ps

Pin-Assignment and Synthesis

- Completed pin assignment from Table 11.1.
- Demonstrate your working stopwatch to a member of the lab team.